

9.15.2.6. In-field upgrade of ICD functionality

This section details the behavior and requirements for an ICD upgrading from a Short Idle Time (SIT) implementation in Matter 1.3 or earlier, to a Long Idle Time (LIT) capable implementation in Matter 1.4 or later. The goal is for an ICD-capable client to detect that the device has been upgraded and is now LIT-capable, and for the client to automatically register with the device and enable the enhanced power savings mode of LIT operation.

Pre-Upgrade State

- The device is running a version of Matter that does not support the LIT ICD feature, either:
 - an older Matter version that did not define the LIT ICD feature yet (Matter 1.3 or earlier)
 - or the device did not implement optional support for the LIT ICD feature (Matter 1.4 or later)
- The device operates as a SIT ICD (i.e., any non-LIT device, including devices running Matter 1.3 or earlier).
- A client (with ICD support) exists on the network, but has not registered with the ICD via the ICD Management cluster.

Post-Upgrade State

- The device upgrades to Matter 1.4 or later and is now LIT capable or otherwise enables LIT support after initial deployment.
- The device now supports the Long Idle Time feature within the ICD Management cluster.

Client Behavior and Discovery

- Following the upgrade, a client will see the ICD Management cluster appear, signaling that the device is now LIT (Long-Idle-Time) capable. This is due to new clusters and attributes becoming available post-upgrade.
- To identify the transition from SIT to LIT, a client SHOULD detect the following:
 - The [ConfigurationVersion](#), when present, increases from its prior value. This SHOULD trigger the client to query for the following device capabilities:
 - The presence of the [ICD Management Cluster](#) either via a read, wildcard subscription, or other mechanisms.
 - The [LongIdleTimeSupport](#) feature within the ICD Management cluster is set to **1**.
 - The ICD discovery TXT key in the DNS-SD record is present.

Handling Subscription Loss in ICD Upgrades

If a client loses its subscription (even with server-side [persistent subscriptions](#)), the client SHOULD:

- Monitor for attribute changes, which the server will report if the subscription is active.
- If no reports are received (indicating a lost subscription), the client SHOULD resubscribe to the ICD Management cluster to detect changes in critical attributes (such as LongIdleTimeSupport, ConfigurationVersion, etc.).